

- 2 Tumours may be described as benign or malignant. Malignant tumours can lead to greater complications for the person with the tumour.

Early identification of tumours, particularly malignant tumours, is important for effective treatment. Benign tumour cells and malignant tumour cells can look similar when viewed using a light microscope.

Some scientists wanted to develop a diagnostic test to identify tumour cells as benign or malignant. The scientists investigated whether the diameter of the cell nucleus could be used to identify the type of tumour cell as benign or malignant.

The scientists used a light microscope with a calibrated eyepiece graticule to measure the diameter of the nuclei of stained tumour cells.

Fig. 2.1 shows a photomicrograph of stained tumour cells viewed using a light microscope.

An eyepiece graticule was placed across the nucleus of one of the tumour cells.

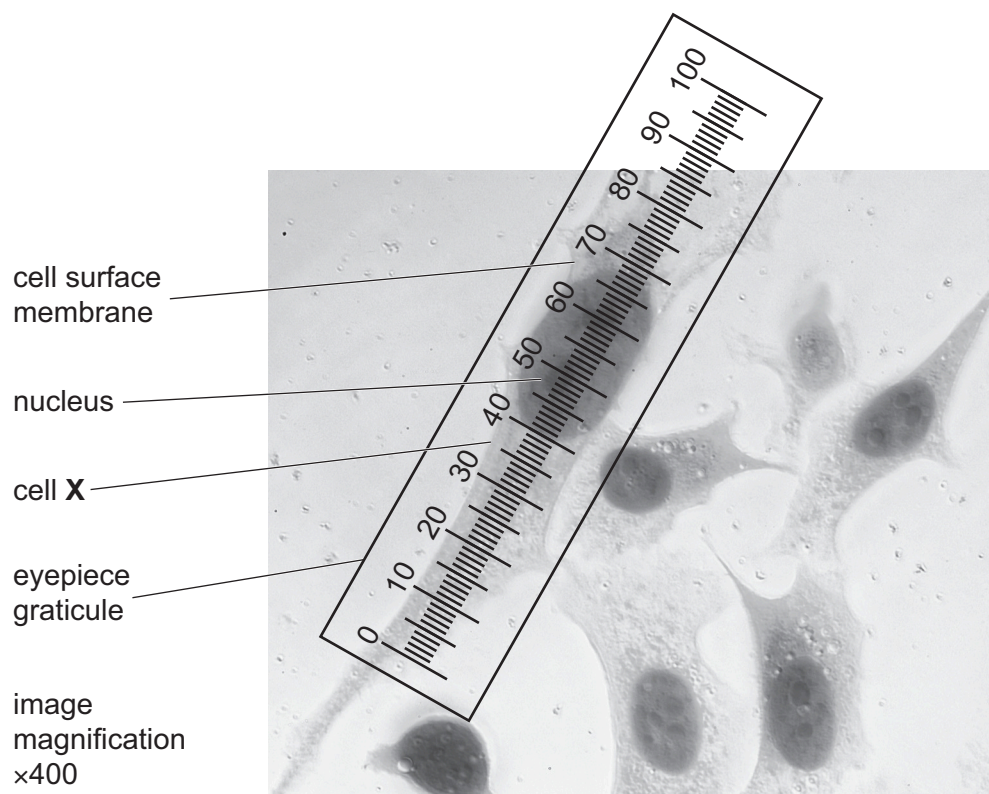


Fig. 2.1

The calibration of the eyepiece graticule scale is:

one eyepiece graticule division = 320 nm



- (a) Use the calibration of the eyepiece graticule scale to calculate the actual diameter of the nucleus of cell **X**, shown in Fig. 2.1.

Show your working and state your answer in μm .

actual diameter of the nucleus of cell **X** = μm [3]

- (b) For the investigation into whether the diameter of the nucleus could be used to identify the type of tumour cell, tumours of the thyroid gland (an endocrine gland) were used.

For the diagnostic test the scientists carried out a procedure using two different stains.

- step 1 50 people diagnosed with benign tumours and 24 people diagnosed with malignant tumours were selected.
- step 2 A sample of tumour cells was removed from the thyroid gland of each person.
- step 3 The cells were stained with either Papanicolaou stain (Pap) or haematoxylin and eosin stain (H&E).
- step 4 Samples were viewed using a light microscope with a magnification of $\times 400$.
- step 5 In each sample, the diameters of 100 nuclei were measured.
- step 6 The measurements were made by one of the scientists who did not know the type of tumour cell they were measuring.



Table 2.1 shows the results of the investigation.

Table 2.1

category	mean nuclear diameter/ μm	standard deviation/ μm	standard error/ μm
total benign cells (50 people) 5000 cells:	7.3	± 0.8	± 0.011
• Pap-stained cells (41 people) 4100 cells	7.3	± 0.7	± 0.011
• H&E-stained cells (9 people) 900 cells	7.2	± 1.0	± 0.033
total malignant cells (24 people) 2400 cells:	9.0	± 0.6	± 0.122
• Pap-stained cells (18 people) 1800 cells	9.0	± 0.7	± 0.016
• H&E-stained cells (6 people) 600 cells	8.8	± 0.5	± 0.020

- (i) Suggest how the scientists can standardise the method of measuring the diameter of the 100 nuclei in step 5, so that valid comparisons can be made between benign and malignant tumour cells.

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- (ii) Identify **one** variable that the scientists have standardised in step 1 to step 6, **other than** details of the method of measurement of the nuclei.

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- (iii) The scientists used a *t*-test to compare the mean nuclear diameter of the Pap-stained cells and the H&E-stained cells in Table 2.1.

State a null hypothesis for the *t*-test.

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- (iv) The t -value the scientists calculated had a probability (p) value greater than 0.25 ($p > 0.25$).

State **one** conclusion that can be made about the effect of using the two stains, Pap and H&E, on the mean nuclear diameter of the cells.

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- (v) State **and** explain **one** conclusion that can be made from the data in Table 2.1 when comparing the results for the 5000 benign cells and the 2400 malignant cells.

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- (vi) At the end of the investigation, the scientists evaluated their procedure and results. They identified some disadvantages of using the procedure as a diagnostic test.

Suggest **and** explain **two** disadvantages of this procedure as a diagnostic test.

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..... [2]

[Total: 11]